Right for the Right Concept: Revising Neuro-Symbolic Concepts by Interacting with their Explanations

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Abstract

Most explanation methods in deep learning map importance estimates for a model's prediction back to the original input space. These "visual" explanations are often insufficient, as the model's actual concept remains elusive. Moreover, without insights into the model's semantic concept, it is difficult —if not impossible— to intervene on the model's behavior via its explanations, called Explanatory Interactive Learning. Consequently, we propose to intervene on a Neuro-Symbolic scene representation, which allows one to revise the model on the semantic level, e.g. "never focus on the color to make your decision". We compiled a novel confounded visual scene data set, the CLEVR-Hans data set, capturing complex compositions of different objects. The results of our experiments on CLEVR-Hans demonstrate that our semantic explanations, i.e. compositional explanations at a per-object level, can identify confounders that are not identifiable using "visual" explanations only. More importantly, feedback on this semantic level makes it possible to revise the model from focusing on these factors.

1. Introduction

Machine learning models may show Clever-Hans like moments when solving a task by learning the "wrong" thing, *e.g.* making use of confounding factors within a data set. Unfortunately, it is not easy to find out whether, say, a deep neural network is making Clever-Hans-type mistakes because they are not reflected in the standard performance measures such as precision and recall. Instead, one looks at their explanations to see what features the network is actually using [23]. By interacting with the explanations, one may even fix Clever-Hans like moments [43, 50, 47, 44].

This Explanatory Interactive Learning (XIL), however, very much depends on the provided explanations. Most explanation methods in deep learning map importance estimates for a model's prediction back to the original input space [46, 49, 48, 45, 7]. This is somewhat reminiscent of a child who points towards something but cannot articulate why something is relevant. In other words, "visual" explanations are insufficient if a task requires a concept-level understanding of a model's decision. Without knowledge about and symbolic access to the concept level, it remains difficult—if not impossible—to fix Clever-Hans behavior.

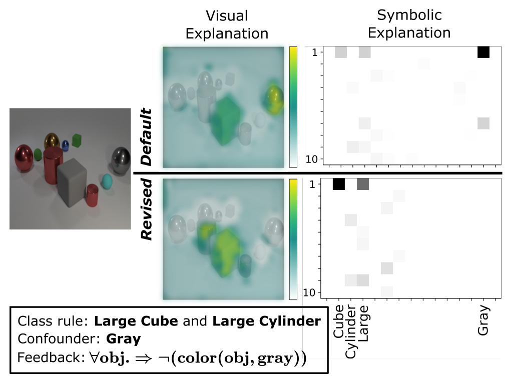

Figure 1: **Neuro-Symbolic explanations are needed to revise deep learning models from focusing on irrelevant features via global feedback rules.**

To illustrate this, consider the classification task depicted in Fig. 1. It shows a complex scene consisting of objects, which vary in position, shape, size, material, and color. Now, assume that scenes belonging to the true class show a large cube and a large cylinder. Unfortunately, during training, our deep network only sees scenes with large, gray cubes. Checking the deep model's decision process using visual explanations confirms this: the deep model has learned to largely focus on the gray cube to classify scenes to be positive. An easy fix would be to provide feedback in the form of "never focus on the color to make your decision" as it would eliminate the confounding factor. Unfortunately, visual explanations do not allow us direct access to the semantic level— they do not tell us that "the color gray

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is an important feature for the task at hand" and we cannot provide feedback at the symbolic level.

Triggered by this, we present the first Neuro-Symbolic XIL (NeSy XIL) approach that is based on decomposing a visual scene into an object-based, symbolic representation and, in turn, allows one to compute and interact with neuro-symbolic explanations. We demonstrate the advantages of NeSy XIL on a newly compiled, confounded data set, called CLEVR-Hans. It consists of scenes that can be classified based on specific combinations of object attributes and relations. Importantly, CLEVR-Hans encodes confounders in a way so that the confounding factors are not separable in the original input space, in contrast to many previous confounded computer vision data sets.

To sum up, this work makes the following contributions: (i) We confirm empirically on our newly compiled confounded benchmark data set, CLEVR-Hans, that Neuro-Symbolic concept learners [31] may show Clever-Hans moments, too. (ii) To this end, we devise a novel Neuro-Symbolic concept learner, combining Slot Attention [28] and Set Transformer [24] in an end-to-end differentiable fashion. (iii) We provide a novel loss to revise this Clever-Hans behaviour. (iv) Given symbolic annotations about incorrect explanations, even across a set of several instances, we efficiently optimize the Neuro-Symbolic concept learner to be right for better Neuro-Symbolic reasons. (v) Thus we introduce the first XIL approach that works on both the visual and the conceptual level. These contributions are important to make progress towards creating conversational explanations between machines and human users [53, 33]. This is necessary for improved trust development and truly Explanatory Interactive Learning: symbolic abstractions help us, humans, to engage in conversations with one another and to convey our thoughts efficiently, without the need to specify much detail.[1]

2. Related Work on XIL

Our work touches upon Explainable AI, Explanatory Interactive Learning, and Neuro-Symbolic architectures.

Explainable AI (XAI) methods, in general, are used to evaluate the reasons for a (black-box) model's decision by presenting the model's explanation in a hopefully human-understandable way. Current methods can be divided into various categories based on characteristics [55], *e.g.* their level of intrinsicality or if they are based on backpropagation computations. Across the spectrum of XAI approaches, from backpropagation-based [49, 2], to model distillation [41], or prototype-based [25] methods, very often an explanation is created by highlighting or otherwise relating direct input elements to the model's prediction, thus visualizing an explanation at the level of the input space.

Several studies have investigated methods that produce explanations other than these visual explanations, such as multi-modal explanations [36, 54, 40], including visual and logic rule explanations [1, 39]. [32, 27] investigate methods for creating more interactive explanations, whereas [3] focuses on creating single-modal, logic-based explanations. Some recent work has also focused on creating concept-based explanations [18, 59, 9]. None of the above studies, however, investigate using the explanations as a means of intervening on the model.

Explanatory interactive learning (XIL) [43, 47, 50, 44] merges XAI with an active learning setting. It incorporates XAI in the learning process by involving the human-user —interacting on the explanations— in the training loop. More precisely, the human user can query the model for explanations of individual predictions and respond by correcting the model if necessary, providing a slightly improved —but not necessarily optimal— feedback on the explanations. Thus, as in active learning, the user can provide the correct label if the prediction is wrong. In addition, XIL also allows the user to provide feedback on the explanation. This combination of receiving explanations and user interaction is a strong necessity for gaining trust in the model's behavior [50, 44]. XIL can be applied to differentiable as well as non-differentiable models [44].

Neuro-Symbolic architectures [8, 57, 31, 13, 52, 6] make use of data-driven, sub-symbolic representations, and symbol-based reasoning systems. This field of research has received increasing interest in recent years as a means of solving issues of individual subsystems, such as the out-of-distribution generalization problem of many neural networks, by combining the advantages of symbolic and sub-symbolic models. Yi *et al.* [57], for example, propose a Neuro-Symbolic based VQA system based on disentangling visual perception from linguistic reasoning. Each submodule of their system processes different subtasks, *e.g.* their scene parser decomposes a visual scene into an object-based scene representation. Their reasoning engine then uses this decomposed scene representation rather than directly computing in the original input space. An approach that also relates to the work of Lampert *et al.* [21, 22].

3. Motivating Example: Color-MNIST

To illustrate the problem setting, we first revert to a well known confounded toy data set. ColorMNIST [17, 42] consists of colored MNIST digits. Within the training set, each number is confounded with a specific color, whereas in the test set, the color association is shuffled or inverted.

A simple CNN model can reach 100% accuracy on the training set, but only 23% on the test set, indicating that the model has learned to largely focus on the color for accurate prediction rather than the digits themselves. Fig. 2 depicts the visual explanation (here created using GradCAM [46])

[1]Source code is available at `https://github.com/ml-research/NeSyXIL`

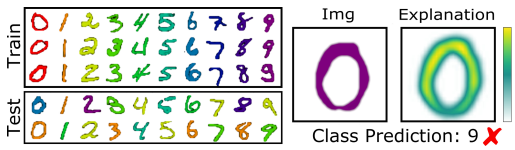

Figure 2: **Visual Explanations for ColorMNIST.** (Left) the general data distribution between train and test split. (Right) a typical visual explanation of a CNN. Notice digit pixels are considered as important for the wrong prediction.

of a zero that is predicted as a nine. Note the zero is colored in the same color as all nines of the training set. From the visual explanation it becomes clear that the model is focusing on the correct object, however why the model is predicting the wrong digit label does not become clear without an understanding of the underlying training data distribution.

Importantly, although the model is wrong for the right reason, it is a non-trivial problem of interacting with the model to revise its decision using XIL solely based on these explanations. Setting a loss term to correct the explanation (*e.g.* [43]) on color channels is as non-trivial and inconvenient as un-confounding the data set with counterexamples [50]. Kim *et al.* [17] describe how to unbias such a data set if the bias is known, using the mutual information between networks trained on separate features of the data set in order for the main network not to focus on bias features. Rieger *el al.* [42] propose an explanation penalization loss similar to [43, 47, 44], focusing on Contextual Decomposition [35] as explanation method. However, the utilized penalization method is task-specific and detached from the model's explanations, resulting in only a slight improvement of a final 31% accuracy (using the inverted ColorMNIST setting).

4. Neuro-Symbolic Explanatory Interactive Learning

The Color-MNIST example clearly shows that although the input-level explanations of current XAI methods are an important first step towards true explanations of a model's behavior, much ambiguity in a model's decision process remains. Using XIL on these visual explanations only, it can be difficult to properly intervene on a model. What we require is an understandable, disentangled representation level, which the user can enquire from and intervene on.

Neuro-Symbolic Architecture. For this purpose, we construct an architecture consisting of two modules, a concept embedding and a reasoning module. The concept module's task is to create a decomposed representation of the input space that can be mapped to human-understandable symbols. The task of the reasoning module is to make predictions based on this symbolic representation.

Fig. 3 gives an illustrative overview of our approach, which we formulate more precisely in the following: Given an input image $x_i \in X$, whereby $X := [x_1, ..., x_N] \in \mathbb{R}^{N \times M}$, with X being divided into subsets of N_c classes $\{X_1, ..., X_{N_c}\} \in X$ and with ground-truth class labels defined as $y \in [0,1]^{N \times N_c}$, we have two modules, the concept embedding module, $h(x_i) = \hat{z}_i$, which receives the input sample and encodes it into a symbolic representation, with $\hat{z} \in [0,\ 1]^{N \times D}$. And the reasoning module, $g(\hat{z}_i) = \hat{y}_i$, which produces the prediction output, $\hat{y}_i \in [0,1]^{N \times N_c}$, given the symbolic representation. The exact details of the $g(\hat{z}_i)$ and $h(x_i)$ depend on the specific implementations of these modules, and will be discussed further in sections below.

Retrieving Neuro-Symbolic Explanations. Given these two modules, we can extract explanations for the separate tasks, *i.e.* the more general input representation task and the reasoning task. We write an explanation function in a general notation as $E(m(\cdot),\ o,\ s)$, which retrieves the explanation of a specific module, $m(\cdot)$, given the module's input s, and the module's output if it is the final module or the explanation of the following module if it is not, both summarized as o here. For our approach, we thus have $E^g(g(\cdot),\ \hat{y}_i,\ z_i) =: \hat{e}^g_i$ and $E^h(h(\cdot),\ \hat{e}^g_i,\ x_i) =: \hat{e}^h_i$. These can represent scalars, vectors, or matrices, depending on the given module and output. $\hat{e}^g_i$ represents the explanation of the reasoning module given the final predicted output $\hat{y}_i$, *e.g.* a logic-based rule. $\hat{e}^h_i$ presents the explanation of the concept module given the explanation of the reasoning module $\hat{e}^g_i$, *e.g.* a visual explanation of a learned concept. In this way, the explanation of the reasoning module is passed back to the concept module in order to receive the explanations of the concept module that contribute to the explanation of the reasoning module. This explanation pass is depicted by the gray arrows of Fig. 3. The exact definition of E^g and E^h used in this work are described below.

Revising Neuro-Symbolic Concepts. As we show in our experiments below, also Neuro-Symbolic models are prone to focusing on wrong reasons, *e.g.* confounding factors. In such a case, it is desirable for a user to intervene on the model, *e.g.* via XIL. As errors can result from different modules of the concept learner, the user must create feedback tailored to the individual module that is producing the error. A user thus receives the explanation of a module, *e.g.* $\hat{e}^g_i$, and produces an adequate feedback given knowledge of the input sample, x_i, the true class label, y_i, the model's class prediction $\hat{y}_i$ and possible internal representations, *e.g.* $\hat{z}_i$. For the user to interact with the model, the user's feedback must be mapped back into a representation space of the model.

In the case of creating feedback for a visual explanation, as in [43], [50] and [44], the mapping is quite clear: the user gives visual feedback denoting which regions of the input are relevant and which are not. This "visual" feedback is transferred to the input space in the form of binary image